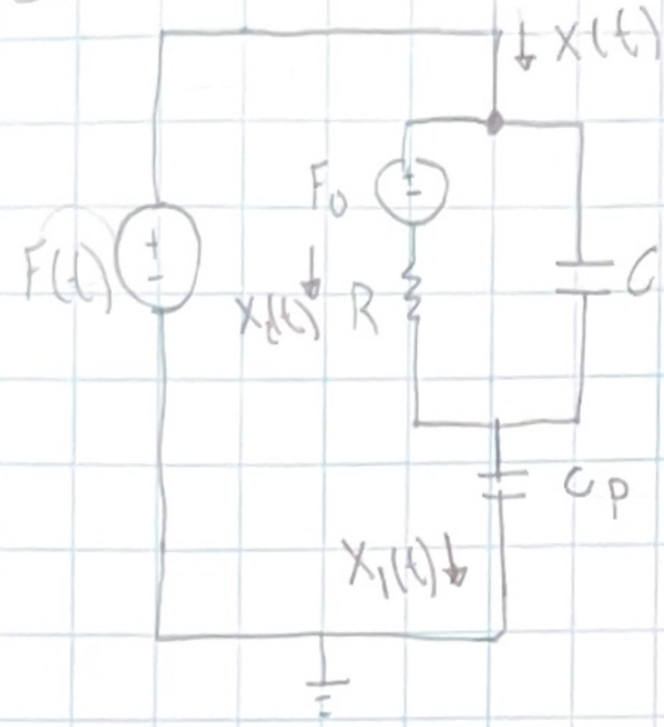
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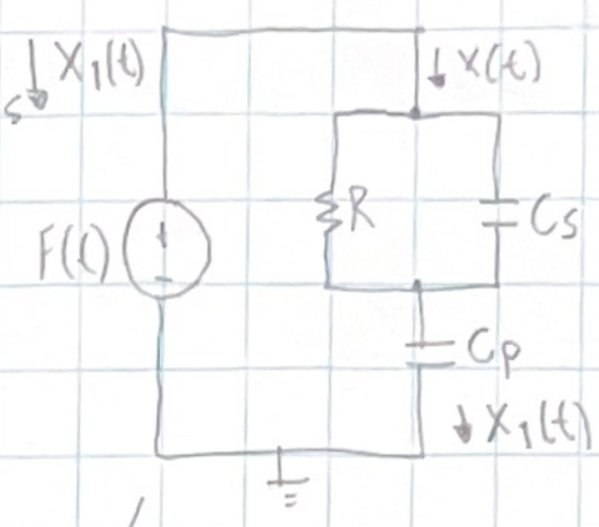
Modelado de Sistemas Fisiologicos.

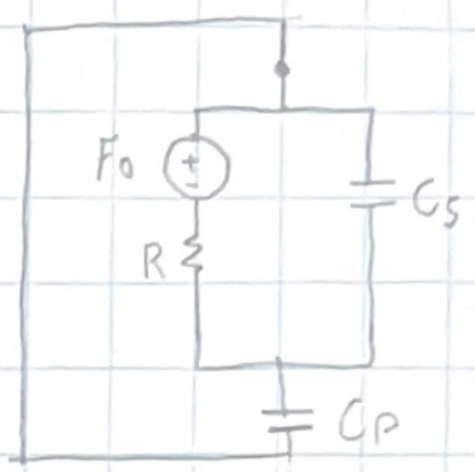
Circuito Electrico



Aplicar Superposición

- Apagamos $F_0(t)$ - Apagamos $F(t)$



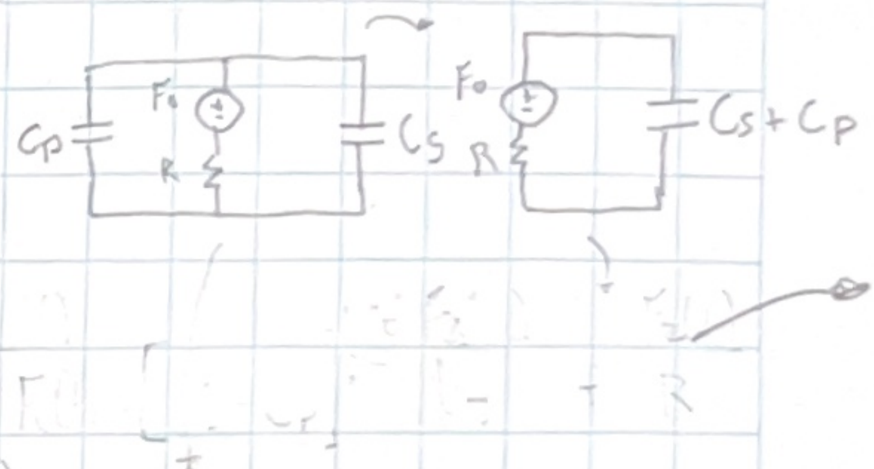


$$X(t) = X_1(t) + X_2(t)$$

$$X_1(t) = C_s \frac{d[F(t) - F_{s1}(t)]}{dt}$$

$$X_2(t) = \frac{F(t) - F_{s1}(t)}{R}$$

$$X(t) = C_p \frac{dF_{s1}(t)}{dt}$$



$$C_p \frac{dF_{s1}(t)}{dt} = C_s \frac{d[F(t) - F_{s1}(t)]}{dt} + \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p F_{s1}(s) = C_s[F(s) - F_{s1}(s)] + \frac{F(s)}{R} - \frac{F_{s1}(s)}{R}$$

$$F_{s1}(s)\left[C_p + C_s + \frac{1}{R}\right] = F(s)\left[C_s + \frac{1}{R}\right]$$

$$F_{s1}(s)\left[\frac{RC_p + RC_s + 1}{R}\right] = F(s)\left[\frac{C_sR + 1}{R}\right]$$

$$F_{s1}(s) = \left[\frac{\frac{C_sR + 1}{R}}{\frac{RC_p + RC_s + 1}{R}}\right] F(s) \qquad F_{s1}(s) = \frac{C_sR + 1}{RC_p + RC_s + 1} \cdot F(s)$$

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Modelado de Sistemas Fisiologicos.

$$-\frac{F_0(t)}{R} - [C_s + C_p]\frac{dF_0(t)}{dt} = [C_s + C_p]\frac{dF_{s2}(t)}{dt}$$

$$F_0(t)\left[-\frac{1}{R} - \frac{d[C_s + C_p]}{dt}\right] = F_{s2}(t)\frac{d[C_s + C_p]}{dt}$$

$$F_0(s)\left[-\frac{1}{R} - [C_s + C_p]\right] = F_{s2}(s)[C_s + C_p]$$

$$F_s(s)\propto\left[-\frac{1}{R} - C_s - C_p\right] = F_{s2}(s)[C_s + C_p]$$

$$F_{s2}(s) = \left[\frac{-1 - C_sR - C_pR}{C_pR + RC_s + 1}\right]F_s(s) \qquad F_{s2}(s) = \left[\frac{-1 - C_sR - C_pR}{RC_s + RC_p}\right]F_s(s)$$

$$\frac{F_s(s)}{F(s)} = \frac{C_sR_s + 1 - \alpha}{R(C_p + C_s)s + 1}$$